Description of Course CSE 316

PART A: General Information

1	**Course Title**	: MICROPROCESSORS, MICROCONTROLLERS, AND EMBEDDED SYSTEMSSESSIONAL
2	**Type of Course**	: SESSIONAL
3	**Offered to**	: DEPARTMENT OF CSE
4	**Pre-requisite Course(s)**	: NONE

PART B: Course Details

1. Course Content (As approved by the Academic Council)

Sessional based on CSE 315 including assembly language programming and experiments using microprocessor/microcontroller.

2. Course Objectives

The students are expected to:

i. Understandand adapt Assembly language
ii. Implement hardware systems through interfacing different computing and memory components
iii. Design and develop digital systems with a microcontroller, sensors, and/or actuators

3. Knowledge required

Technical

- System design and development

Analytical

- Logic design

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4. Course Outcomes (COs)

CO No.	CO Statement After undergoing this course, students should be able to:	Corresponding PO(s)*	Domains and Taxonomy level(s)**	Delivery Method(s) and Activity(-ies)	Assessment Tool(s)
CO1	**Understand** and **apply** Assembly language	PO1 and PO2	C3	Lecture, Demonstration, and hands-on	Assignments and Quiz
CO2	**Implement** hardware systems through interfacing different computing and memory components.	PO3 and PO4	C3, A5	Lecture, Demonstration, and hands-on	Assignments and Quiz
CO3	**Design** and **develop** digital systems with a microcontroller, sensors, and/or actuators.	PO5 and PO6	C6, P7	Lecture, Demonstration, and hands-on	Projectand Quiz

***Program Outcomes (POs)**

PO1: Engineering knowledge; PO2: Problem analysis; PO3: Design/development of solutions; PO4: Investigation; PO5: Modern tool usage; PO6: The engineer and society; PO7: Environment and sustainability; PO8: Ethics; PO9: Individual work and teamwork; PO10: Communication; PO11: Project management and finance; PO12: Life-long learning.

****Domains**

C-Cognitive: C1: Knowledge; C2: Comprehension; C3: Application; C4: Analysis; C5: Synthesis; C6: Evaluation
A-Affective: A1: Receiving; A2: Responding; A3: Valuing; A4: Organizing; A5: Characterizing
P-Psychomotor: P1: Perception; P2: Set; P3: Guided Response; P4: Mechanism; P5: Complex Overt Response; P6: Adaptation; P7: Organization

5. Mapping of Knowledge Profile, Complex Engineering Problem Solving and Complex Engineering Activities

COs	K1	K2	K3	K4	K5	K6	K7	K8	P1	P2	P3	P4	P5	P6	P7	A1	A2	A3	A4	A5
CO1		√	√	√	√	√	√		√	√	√	√				√			√	
CO2		√	√	√	√	√	√		√	√	√	√			√	√	√			
CO3		√	√	√	√	√	√	√	√	√	√				√	√		√		√

K-Knowledge Profile:

K1: A systematic, theory-based understanding of the natural sciences applicable to the discipline; **K2:** Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline; **K3:** A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline; **K4:** Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline; **K5:** Knowledge that supports engineering design in a practice area; **K6:** Knowledge of engineering practice (technology) in the practice areas in the engineering discipline; **K7:** Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability; **K8:** Engagement with selected knowledge in the research literature of the discipline

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P-Range of Complex Engineering Problem Solving:

P1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach; **P2:** Involve wide-ranging or conflicting technical, engineering and other issues; **P3:** Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models; **P4:** Involve infrequently encountered issues; **P5:** Are outside problems encompassed by standards and codes of practice for professional engineering; **P6:** Involve diverse groups of stakeholders with widely varying needs; **P7:** Are high level problems including many component parts or sub-problems

A-Range of Complex Engineering Activities:

A1: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies); **A2:** Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues; **A3:** Involve creative use of engineering principles and research-based knowledge in novel ways; **A4:** Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation; **A5:** Can extend beyond previous experiences by applying principles-based approaches

6. Lecture/ Activity Plan (Tentative)

Week	Lecture Topics	Corresponding CO(s)
Week 1-2	Introduction	-
Week 3-4	Online on 8086 Assembly Demonstration on ATMega32	CO1
Week 5-6	Hardware Experiment on microcontroller: interfacing LED Matrix with ATMega32	CO2
Week 7-8	Term Project Proposal	CO2 and CO3
Week 9-10	Hardware Experiment on microcontroller: ADC and LCD Display in ATMega32	CO2
Week 11-12	Project Update 1	CO2 and CO3
Week 13-14	Final Project Presentation	CO2 and CO3

7. Assessment Strategy (Tentative)

- Class Attendance: Class attendance will be recorded in every class.
- Assignments and Project: There will be an assignment, two experiments and one Term Project.

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8. Distribution of Marks (Tentative)

Attendance:	10%
Assignments:	20%
Experiments:	30%
Term Project:	40%
Total:	100%

9. Textbook/ Reference

a. Embedded Systems Architecture by Daniele Lacamera
https://www.amazon.com/Embedded-Systems-Architecture-architectural-pragmatic/dp/1788832507/

b. Embedded Systems: Introduction to Arm® Cortex™-M Microcontrollers by Jonathan W Valvano
https://www.amazon.com/Embedded-Systems-Introduction-CortexTM-M-Microcontrollers/dp/1477508996

c. Embedded Systems with ARM Cortex-M Microcontrollers by Yifeng Zhu
https://web.eece.maine.edu/~zhu/book/

d. Edward A. Lee and Sanjit A. Seshia, Introduction to Embedded Systems, A CyberPhysical Systems Approach
https://ptolemy.berkeley.edu/books/leeseshia/releases/LeeSeshia_DigitalV2_2.pdf

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